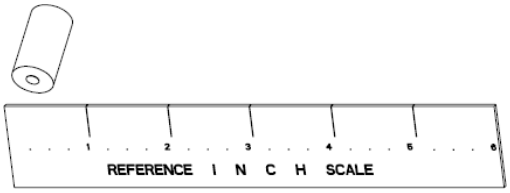
INCH-POUND
MIL-PRF-32271/8A
03 JUNE 2025
SUPERCEDING
MIL-PRF-32271/8
30 OCTOBER 2008

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-32271.



See NOTES for nominal dimensions; see applicable drawing for dimensional requirements.

FIGURE 1. General view, M32271/8 battery shape

For clarity, MIL-PRF-32271 tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32271/8A tables are labeled with lower case Roman numerals (e.g. ix).

REQUIREMENTS (see 1.2 for Type, Class, and Features descriptions):

Type	I
Class	1
Features	A
PIN	M32271/8-11A

Distribution Statement A. Approved for public release: distribution is unlimited.
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Specification requirements: The following requirements of MIL-PRF-32271, identified therein as "when specified", are applicable as indicated below:

TABLE i. *"When specified" requirements*

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/8-)
			11A
Parallel discharges	*Tables II, III, IV, IX, X, XI, & XII*		*No*
Additional material requirements	*3.3.5*	*4.6.1.11*	*No*
Parallel cell arrangements	*3.4*		*No*
Connectors	*3.4.5*	*4.6.1.7*	*No*
Battery charger connection	*3.4.5.1*	*4.6.1.7.1*	*No*
Connection integrity	*3.4.5.2*		
Static	*3.4.5.2*	*4.6.1.7.2*	*No*
Dynamic	*3.4.5.2*	*4.6.1.7.3*	*No*
Terminal integrity	3.4.6		
Socket Strength	*3.4.6.1*	*4.6.1.8.1*	*No*
Terminal Strength	3.4.6.2	4.6.1.8.2	Yes
Complete discharge device	*3.4.7*	*4.6.1.9*	*No*
State of charge device	*3.4.8*	*4.6.1.10*	*No*
State of charge data output	*3.4.8e*	*4.6.1.10*	*No*
Additional design and construction requirements	*3.4.11*	*4.6.1.12*	*No*
Cell charging	*3.5.3*	*4.6.2.3*	*No*
Nail penetration	*3.5.4*	*4.6.2.4*	*No*
Parallel string charge protection	*3.5.6*	*4.6.2.6*	*No*
Cell series string short circuit	*3.5.7*	*4.6.2.7*	*No*
Battery charge protection	*3.5.12*	*4.6.2.12*	*No*
Over-current protection	*3.5.14.1*	*4.6.2.14*	*No*
Over-temperature protection	*3.5.15*	*4.6.2.15*	*No*
Projectile	3.5.16	4.6.2.17	Yes
Additional safety requirements	*3.5.18*	*4.6.2.18*	*No*
Capacity LR	*3.6*	*4.6.4.1.5*	*No*
Capacity LRT	*3.6*	*4.6.4.1.9*	*No*
Capacity IDC	*3.6*	*4.6.4.1.10*	*No*
Surface temperature	*3.6h*	*4.6.4.1*	*No*
Immersion	3.7.8	4.6.5.8	Yes
Watertight integrity	*3.7.9*		
Shallow	*3.7.9*	*4.6.5.9*	*No*
Deep	*3.7.9*	*4.6.5.10*	*No*
Electromagnetic interference	*3.7.10*	*4.6.5.11*	*No*
Electrostatic discharge	*3.7.11*	*4.6.5.12*	*No*
Additional environmental requirements	*3.7.12*	*4.6.5.13*	*No*

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TABLE i. *"When specified" requirements* – Continued

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/8-)
			11A
DO NOT CHARGE! marking	*3.8.1.2b*		*No*
Complete discharge device marking/label	*3.8.4*		*No*
State of charge marking	*3.8.5*		*No*
Simplified battery label	*3.8.7*		*No*
Additional markings	*3.8.8*		*No*

Dimensions, marking and configuration

Battery - Drawing A3315885
Battery Connector – Not Applicable
Mating Connector – Not Applicable
Battery Charger Connector – Not Applicable

Battery voltages (3.4.4 & 4.6.1.5):

Battery open-circuit voltage (3.4.4.1 & 4.6.1.5.1):
Maximum – 6.8 volts
Minimum – 6.0 volts

Battery closed circuit voltage (3.4.4.2 & 4.6.1.5.2):
For capacity and initial voltage delay: Minimum – 4.0 volts
For battery closed circuit voltage test: Minimum – 5.0 volts

Cell closed circuit voltage test (3.4.4.3 & 4.6.1.3): Minimum - 2.5 volts per cell

Maximum weight (3.1 & 4.6.1.6):

0.7 ounces (19.8 g)

Capacity test requirements (3.6 & 4.6.4):

Test	Initial voltage delay Time (MAX) to	Volts	Minimum Capacity		Cut-off Volts
I	1 second	4.0	100 hours		4.0
L	1 second	4.0	75 hours		4.0
H	1 second	4.0	100 hours		4.0
			Storage Period		
			1-Week	4-Week	
IT	1 second	4.0	93 hours	90 hours	4.0
LT	1 second	4.0	67 hours	60 hours	4.0
HT	1 second	4.0	93 hours	90 hours	4.0

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Abuse test pulse discharge capacity requirement (3.5.12 & 4.6.2.12i):

0.42 ampere-hours

METHODS OF EXAMINATION AND TEST:

Cell closed-circuit voltage test: When cells are tested as specified in 4.6.1.3, load each cell with 150 ohms, 4.5 milli-amperes, or 9 milli-watts.

Battery closed-circuit voltage test: When tested as specified in 4.6.1.5.2, load each battery with 298.2 ohms, 4.5 milli-amperes, or 18 milli-watts.

Abuse test, pre-discharge: When tested as specified in 4.6.2.13a, discharge with a load of 4.5 milli-amperes for 50 hours.

Abuse test, pulse discharge: When tested as specified in 4.6.2.13l, discharge with a load of 5.5 milli-amperes for 1 minute followed by 4 milli-amperes for 4 minutes, cycled continuously to 4.0 volts.

Immersion: When tested as specified in 4.6.5.8, apply a load of 1 milli-ampere during the storage and immersion time periods specified.

Capacity tests (4.6.4):

Test	Discharge Rate	Duty Cycle
I	4.5 milli-amperes	Continuous discharge to zero volts, followed by 4.5 milli-ampere forced discharge for 5 minutes.
L	4.5 milli-amperes	Continuous discharge to cut-off voltage
H	4.5 milli-amperes	Continuous discharge to cut-off voltage
IT	4.5 milli-amperes	Continuous discharge to cut-off voltage
LT	4.5 milli-amperes	Continuous discharge to cut-off voltage
HT	4.5 milli-amperes	Continuous discharge to cut-off voltage

NOTES:

(This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This specification covers non-rechargeable lithium batteries of the non-reserve type composed of electrochemical cells using lithium metal in the anode. They are intended for use in military communications and electronics equipment.

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Nominal ratings[1]: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes.

Battery PIN:	**M32271/8-11A (Li/MnO_2)**
Overall Dimensions:	1.3 in L x 0.6525 DIA
MAX Weight:	0.7 ounces (19.8 grams)
Voltage Range:	4.0-6.8
Nominal Capacity: (in ampere-hours)	0.45 Amp-hrs
Nominal Energy: (in watt-hours)	2.3 Watt-hrs
MAX rated power output:	27 milli-watts
MAX continuous load rating:	4.5 milli-amperes
MAX pulse load rating:	5.5 milli-amperes
Instantaneous trip rating:	Not applicable
Operating temperature range:	-4 to 130ºF (-20 to 55ºC)
Storage temperature range:	-40 to 160ºF (-40 to 71ºC)
MAX abusive temperature: (non-operating)	195ºF (91ºC)

Other data:

Military Type Designations. The military type designation that relates to the PIN covered by this specification sheet is as follows: The BA-5372()/U designation has been used for the Type I PIN.

This battery was exempt from the transportation requirements of 49 CFR 173.185 until January 2008; now it is subject to the test requirements therein beginning in October 2009. Class 9 shipping labels typically won't be needed; consult with 49 CFR 173.185 for package quantity/size limitations. It contains less than 0.5 of lithium per cell and less than 1.0 gram of lithium per battery. It should be noted that all non-rechargeable lithium batteries are restricted from shipment as cargo aboard passenger aircraft within, entering, or leaving the US.

This battery does not have a complete discharge device. It is considered Non-Hazardous Solid Waste per the Resource Conservation and Recovery Act (RCRA). However, it is considered Hazardous Waste in states using Bioassay rules. Check with your local disposal office for rules that apply.

The US Army CECOM Life Cycle Management Command (LCMC) publishes guidelines for the design of battery compartments for devices using the Type I battery covered by this specification sheet: Technical Bulletin 43-6135. Please contact the CECOM LCMC Directorate for Safety at usarmy.APG.cecom.mbx.amsel-sf-sec-cecom-lcmc--ds@army.mil for further information.

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery may require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

[1] Capacity and energy delivered by a battery are significantly affected by usage conditions, such as temperature and loads applied. If you have any questions about use of this battery in a particular device or circumstance, please visit the following web site (contact info is posted): https://battery.army.mil.

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Navy – SH
Air Force – 99
DLA – GS

Review activities:
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Preparing activity:
Army – CR
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